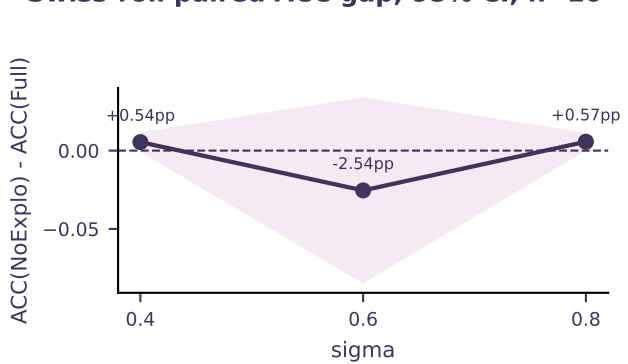

Warm-pink x: P2_Full; blue o: P2_NoExploration; black *: eccentric decoys. Post-hoc colors differ from main-text method identity colors. Spiral (near-inert) and hourglass (negative-control geometry) full results are reported in Supplementary Table X; all 180 shards and nine datasets are released in the repository.

Post-hoc Mechanism Validation — not part of pre-registered P1-P4